# EXPERIMENT #50

Timestamp: 20260330_162533 | Environment: Local

## 1. Experiment Parameters & Network Statistics

| Parameter | Value |
|---|---|
| Number of sticks (N) | 30 |
| Stick length (a) | 0.65 |
| Domain size | 1.0 x 1.0 |
| Source region | x < 0.1 |
| Drain region | x > 0.9 |
| Total intersections | 198 |
| Has percolation | [OK] Yes |
| # Percolating paths | 1 |
| Nodes before pruning | 198 |
| Nodes after pruning | 198 |
| Edges before pruning | 366 |
| Edges after pruning | 366 |

## 2. Network Visualizations

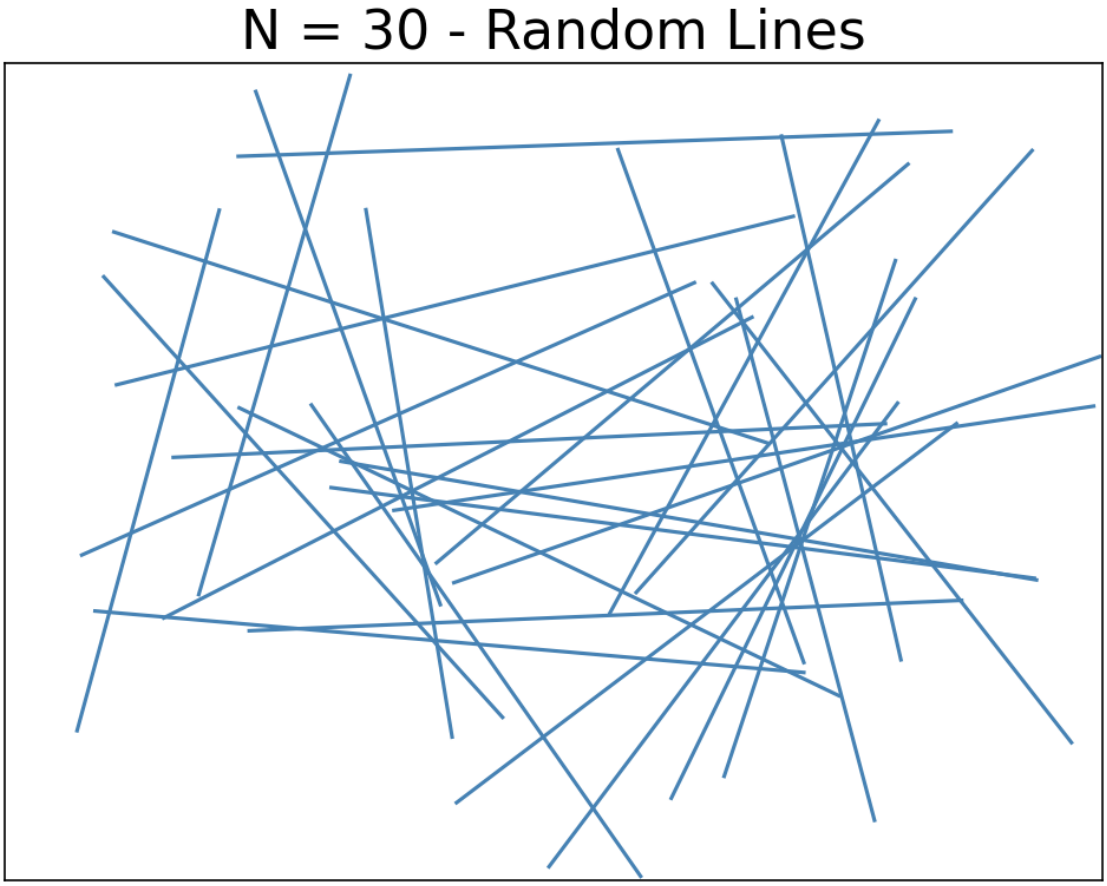



Fig. -- 01 random lines N30

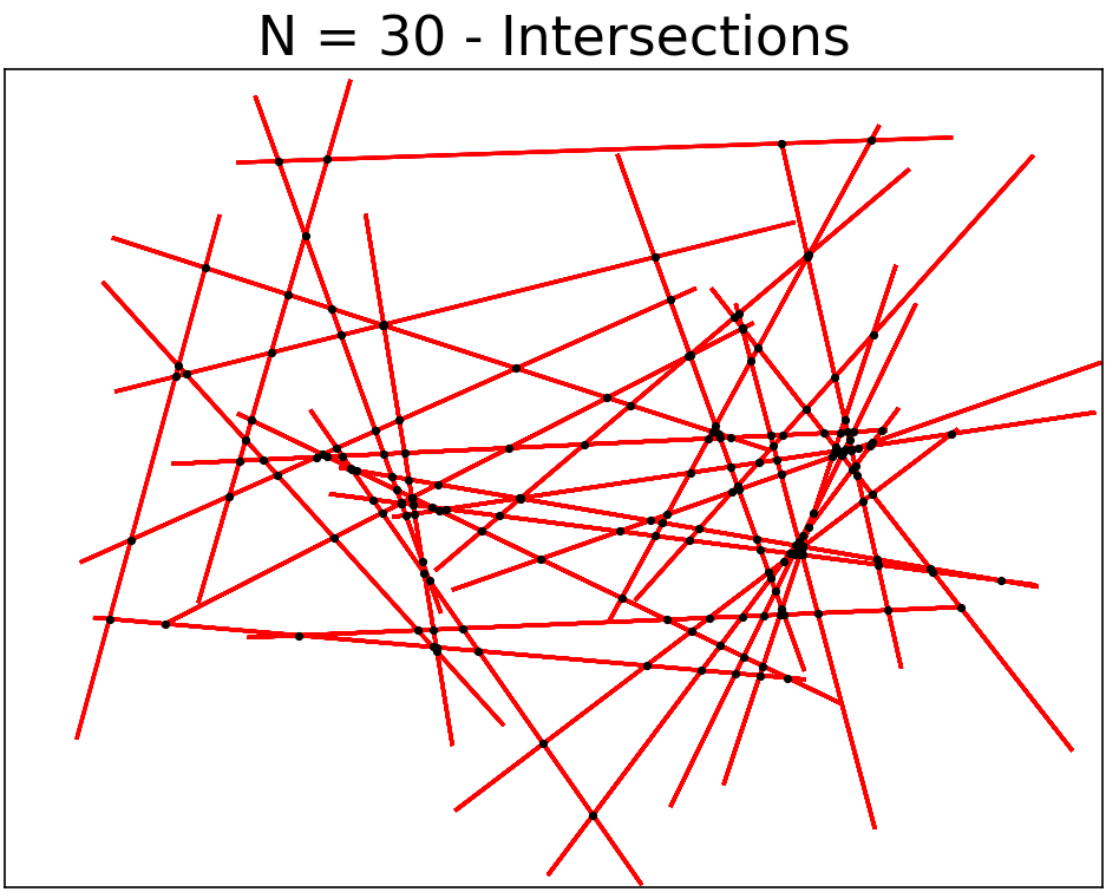



Fig. -- 02 intersections N30

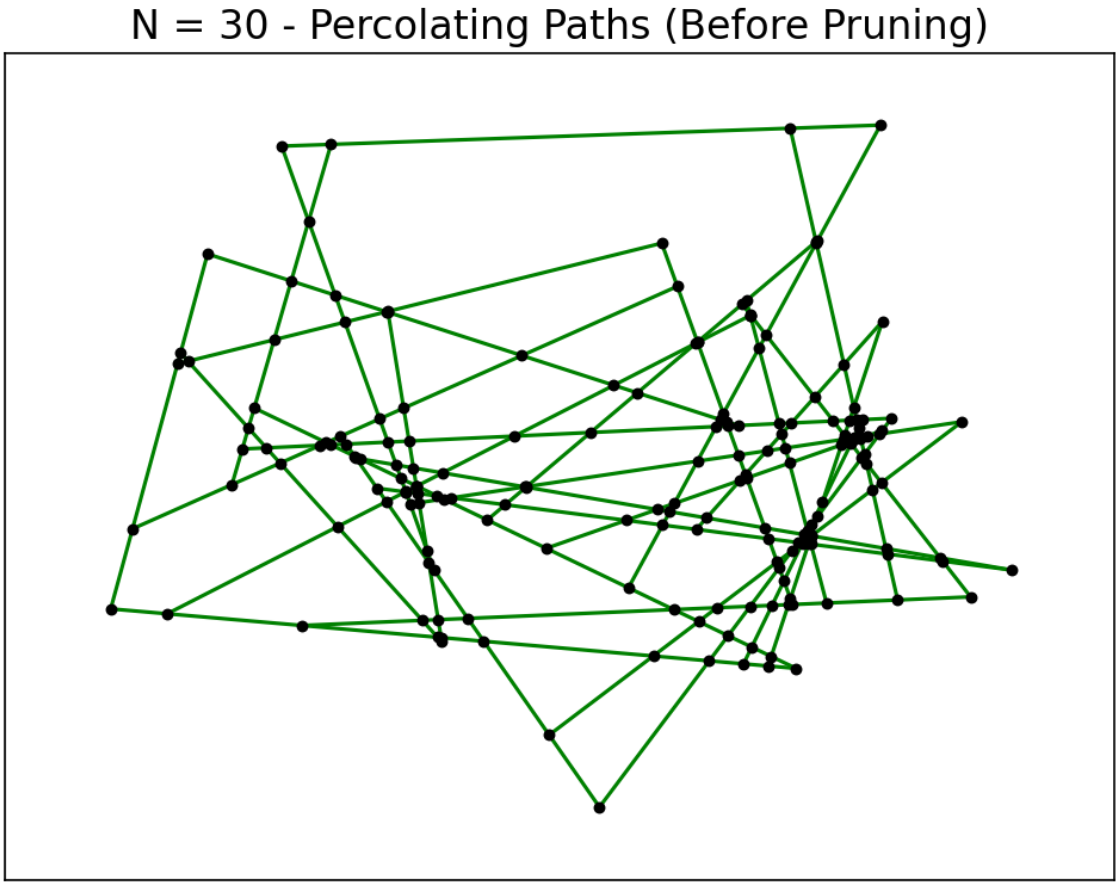



Fig. -- 03 percolating paths before pruning N30

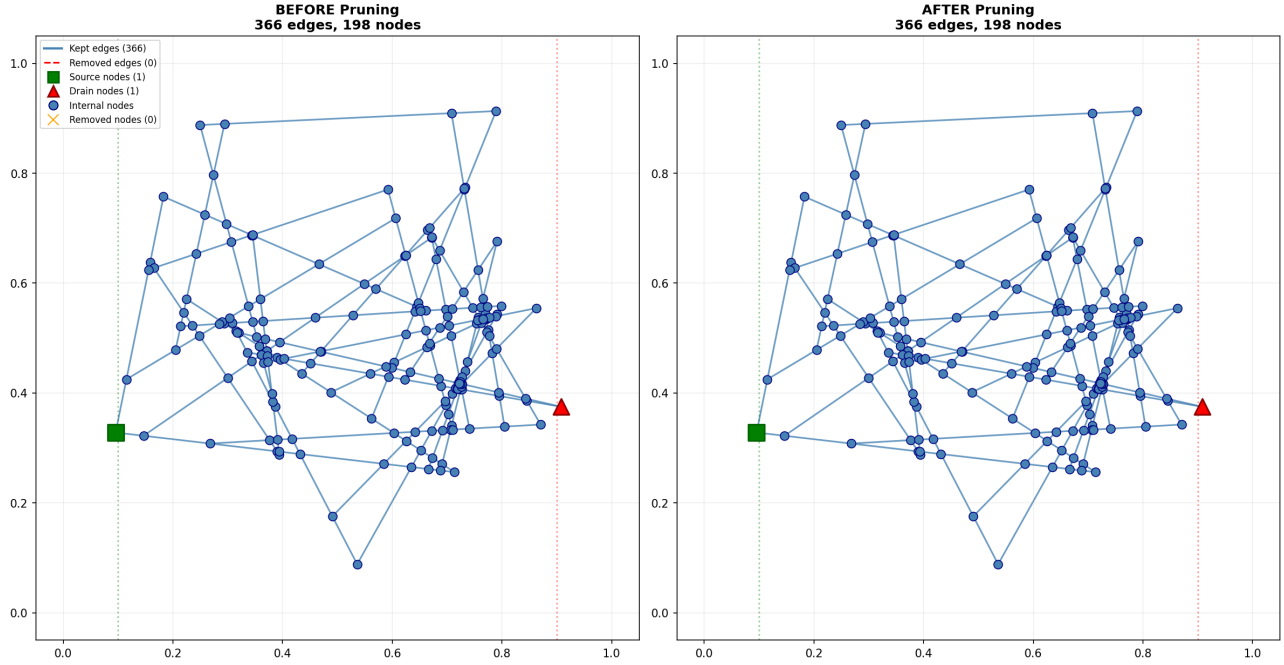



Fig. -- 04 before after pruning N30

## 3. Edge Resistances (Pruned Network)

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 0 | 6 | 2 | 1.0000 | 1.0000 | 0.014639 |
| 1 | 2 | 1 | 1.0000 | 1.0000 | 0.018168 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 2 | 1 | 5 | 1.0000 | 1.0000 | 0.006100 |
| 3 | 5 | 10 | 1.0000 | 1.0000 | 0.005777 |
| 4 | 10 | 8 | 1.0000 | 1.0000 | 0.012105 |
| 5 | 8 | 4 | 1.0000 | 1.0000 | 0.004840 |
| 6 | 4 | 9 | 1.0000 | 1.0000 | 0.002974 |
| 7 | 9 | 12 | 1.0000 | 1.0000 | 0.009421 |
| 8 | 12 | 3 | 1.0000 | 1.0000 | 0.011754 |
| 9 | 3 | 11 | 1.0000 | 1.0000 | 0.002111 |
| 10 | 11 | 7 | 1.0000 | 1.0000 | 0.004868 |
| 11 | 7 | 0 | 1.0000 | 1.0000 | 0.013326 |
| 12 | 70 | 64 | 1.0000 | 1.0000 | 0.013645 |
| 13 | 64 | 63 | 1.0000 | 1.0000 | 0.009653 |
| 14 | 63 | 56 | 1.0000 | 1.0000 | 0.020386 |
| 15 | 56 | 68 | 1.0000 | 1.0000 | 0.015489 |
| 16 | 68 | 62 | 1.0000 | 1.0000 | 0.012974 |
| 17 | 62 | 67 | 1.0000 | 1.0000 | 0.013493 |
| 18 | 67 | 42 | 1.0000 | 1.0000 | 0.010813 |
| 19 | 42 | 59 | 1.0000 | 1.0000 | 0.009906 |
| 20 | 59 | 69 | 1.0000 | 1.0000 | 0.008589 |
| 21 | 69 | 57 | 1.0000 | 1.0000 | 0.022601 |
| 22 | 57 | 58 | 1.0000 | 1.0000 | 0.017484 |
| 23 | 58 | 0 | 1.0000 | 1.0000 | 0.011023 |
| 24 | 0 | 66 | 1.0000 | 1.0000 | 0.002303 |
| 25 | 66 | 29 | 1.0000 | 1.0000 | 0.019014 |
| 26 | 29 | 14 | 1.0000 | 1.0000 | 0.024351 |
| 27 | 14 | 65 | 1.0000 | 1.0000 | 0.023924 |
| 28 | 65 | 61 | 1.0000 | 1.0000 | 0.015552 |
| 29 | 61 | 60 | 1.0000 | 1.0000 | 0.005710 |
| 30 | 74 | 71 | 1.0000 | 1.0000 | 0.025749 |
| 31 | 71 | 1 | 1.0000 | 1.0000 | 0.021697 |
| 32 | 1 | 81 | 1.0000 | 1.0000 | 0.033766 |
| 33 | 81 | 76 | 1.0000 | 1.0000 | 0.023944 |
| 34 | 76 | 77 | 1.0000 | 1.0000 | 0.026742 |
| 35 | 77 | 78 | 1.0000 | 1.0000 | 0.022957 |
| 36 | 78 | 82 | 1.0000 | 1.0000 | 0.018914 |
| 37 | 82 | 72 | 1.0000 | 1.0000 | 0.014041 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 38 | 72 | 15 | 1.0000 | 1.0000 | 0.026640 |
| 39 | 15 | 73 | 1.0000 | 1.0000 | 0.034508 |
| 40 | 73 | 30 | 1.0000 | 1.0000 | 0.029690 |
| 41 | 30 | 80 | 1.0000 | 1.0000 | 0.035124 |
| 42 | 80 | 75 | 1.0000 | 1.0000 | 0.007524 |
| 43 | 75 | 79 | 1.0000 | 1.0000 | 0.010182 |
| 44 | 57 | 101 | 1.0000 | 1.0000 | 0.005117 |
| 45 | 101 | 105 | 1.0000 | 1.0000 | 0.005041 |
| 46 | 105 | 106 | 1.0000 | 1.0000 | 0.000123 |
| 47 | 106 | 103 | 1.0000 | 1.0000 | 0.007019 |
| 48 | 103 | 99 | 1.0000 | 1.0000 | 0.001116 |
| 49 | 99 | 45 | 1.0000 | 1.0000 | 0.002480 |
| 50 | 45 | 102 | 1.0000 | 1.0000 | 0.000494 |
| 51 | 102 | 100 | 1.0000 | 1.0000 | 0.012852 |
| 52 | 100 | 104 | 1.0000 | 1.0000 | 0.004110 |
| 53 | 104 | 71 | 1.0000 | 1.0000 | 0.000522 |
| 54 | 71 | 2 | 1.0000 | 1.0000 | 0.003529 |
| 55 | 58 | 108 | 1.0000 | 1.0000 | 0.006461 |
| 56 | 108 | 113 | 1.0000 | 1.0000 | 0.007403 |
| 57 | 113 | 3 | 1.0000 | 1.0000 | 0.013241 |
| 58 | 3 | 114 | 1.0000 | 1.0000 | 0.022884 |
| 59 | 114 | 110 | 1.0000 | 1.0000 | 0.018789 |
| 60 | 110 | 111 | 1.0000 | 1.0000 | 0.027527 |
| 61 | 111 | 109 | 1.0000 | 1.0000 | 0.021406 |
| 62 | 109 | 107 | 1.0000 | 1.0000 | 0.013442 |
| 63 | 107 | 112 | 1.0000 | 1.0000 | 0.012092 |
| 64 | 112 | 72 | 1.0000 | 1.0000 | 0.016698 |
| 65 | 72 | 115 | 1.0000 | 1.0000 | 0.004099 |
| 66 | 115 | 17 | 1.0000 | 1.0000 | 0.000831 |
| 67 | 119 | 117 | 1.0000 | 1.0000 | 0.010414 |
| 68 | 117 | 4 | 1.0000 | 1.0000 | 0.010727 |
| 69 | 4 | 99 | 1.0000 | 1.0000 | 0.018541 |
| 70 | 99 | 46 | 1.0000 | 1.0000 | 0.017176 |
| 71 | 46 | 120 | 1.0000 | 1.0000 | 0.023863 |
| 72 | 120 | 122 | 1.0000 | 1.0000 | 0.017678 |
| 73 | 122 | 121 | 1.0000 | 1.0000 | 0.016781 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 74 | 121 | 116 | 1.0000 | 1.0000 | 0.014229 |
| 75 | 116 | 123 | 1.0000 | 1.0000 | 0.020128 |
| 76 | 123 | 118 | 1.0000 | 1.0000 | 0.032088 |
| 77 | 100 | 127 | 1.0000 | 1.0000 | 0.008742 |
| 78 | 127 | 47 | 1.0000 | 1.0000 | 0.009004 |
| 79 | 47 | 5 | 1.0000 | 1.0000 | 0.013639 |
| 80 | 5 | 131 | 1.0000 | 1.0000 | 0.025516 |
| 81 | 131 | 126 | 1.0000 | 1.0000 | 0.020130 |
| 82 | 126 | 125 | 1.0000 | 1.0000 | 0.021885 |
| 83 | 125 | 132 | 1.0000 | 1.0000 | 0.021200 |
| 84 | 132 | 107 | 1.0000 | 1.0000 | 0.016743 |
| 85 | 107 | 128 | 1.0000 | 1.0000 | 0.018094 |
| 86 | 128 | 18 | 1.0000 | 1.0000 | 0.030600 |
| 87 | 18 | 73 | 1.0000 | 1.0000 | 0.041242 |
| 88 | 73 | 32 | 1.0000 | 1.0000 | 0.046060 |
| 89 | 32 | 130 | 1.0000 | 1.0000 | 0.103280 |
| 90 | 130 | 124 | 1.0000 | 1.0000 | 0.014936 |
| 91 | 124 | 129 | 1.0000 | 1.0000 | 0.051060 |
| 92 | 133 | 134 | 1.0000 | 1.0000 | 0.013463 |
| 93 | 134 | 74 | 1.0000 | 1.0000 | 0.040388 |
| 94 | 74 | 6 | 1.0000 | 1.0000 | 0.014639 |
| 95 | 91 | 48 | 1.0000 | 1.0000 | 0.010280 |
| 96 | 48 | 136 | 1.0000 | 1.0000 | 0.018207 |
| 97 | 136 | 140 | 1.0000 | 1.0000 | 0.014267 |
| 98 | 140 | 59 | 1.0000 | 1.0000 | 0.016237 |
| 99 | 59 | 143 | 1.0000 | 1.0000 | 0.128815 |
| 100 | 143 | 101 | 1.0000 | 1.0000 | 0.092509 |
| 101 | 101 | 108 | 1.0000 | 1.0000 | 0.018829 |
| 102 | 108 | 7 | 1.0000 | 1.0000 | 0.017887 |
| 103 | 7 | 142 | 1.0000 | 1.0000 | 0.055468 |
| 104 | 142 | 137 | 1.0000 | 1.0000 | 0.107303 |
| 105 | 137 | 19 | 1.0000 | 1.0000 | 0.023952 |
| 106 | 19 | 33 | 1.0000 | 1.0000 | 0.018677 |
| 107 | 33 | 139 | 1.0000 | 1.0000 | 0.018605 |
| 108 | 139 | 135 | 1.0000 | 1.0000 | 0.035068 |
| 109 | 135 | 138 | 1.0000 | 1.0000 | 0.046711 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 110 | 138 | 75 | 1.0000 | 1.0000 | 0.074232 |
| 111 | 75 | 124 | 1.0000 | 1.0000 | 0.091938 |
| 112 | 124 | 141 | 1.0000 | 1.0000 | 0.334672 |
| 113 | 167 | 165 | 1.0000 | 1.0000 | 0.136437 |
| 114 | 165 | 102 | 1.0000 | 1.0000 | 0.041583 |
| 115 | 102 | 51 | 1.0000 | 1.0000 | 0.013203 |
| 116 | 51 | 8 | 1.0000 | 1.0000 | 0.013152 |
| 117 | 8 | 117 | 1.0000 | 1.0000 | 0.005887 |
| 118 | 117 | 164 | 1.0000 | 1.0000 | 0.103864 |
| 119 | 164 | 171 | 1.0000 | 1.0000 | 0.343937 |
| 120 | 171 | 170 | 1.0000 | 1.0000 | 0.078235 |
| 121 | 170 | 110 | 1.0000 | 1.0000 | 0.373403 |
| 122 | 110 | 169 | 1.0000 | 1.0000 | 0.032961 |
| 123 | 169 | 137 | 1.0000 | 1.0000 | 0.040365 |
| 124 | 137 | 36 | 1.0000 | 1.0000 | 0.001425 |
| 125 | 36 | 22 | 1.0000 | 1.0000 | 0.013326 |
| 126 | 22 | 166 | 1.0000 | 1.0000 | 0.072634 |
| 127 | 166 | 149 | 1.0000 | 1.0000 | 0.025298 |
| 128 | 149 | 168 | 1.0000 | 1.0000 | 0.136342 |
| 129 | 23 | 176 | 1.0000 | 1.0000 | 0.090697 |
| 130 | 176 | 77 | 1.0000 | 1.0000 | 0.049600 |
| 131 | 77 | 150 | 1.0000 | 1.0000 | 0.038713 |
| 132 | 150 | 126 | 1.0000 | 1.0000 | 0.071419 |
| 133 | 126 | 178 | 1.0000 | 1.0000 | 0.068144 |
| 134 | 178 | 174 | 1.0000 | 1.0000 | 0.014087 |
| 135 | 174 | 119 | 1.0000 | 1.0000 | 0.101285 |
| 136 | 119 | 164 | 1.0000 | 1.0000 | 0.093450 |
| 137 | 164 | 9 | 1.0000 | 1.0000 | 0.111617 |
| 138 | 9 | 103 | 1.0000 | 1.0000 | 0.020399 |
| 139 | 103 | 52 | 1.0000 | 1.0000 | 0.026302 |
| 140 | 52 | 177 | 1.0000 | 1.0000 | 0.156905 |
| 141 | 177 | 157 | 1.0000 | 1.0000 | 0.088134 |
| 142 | 157 | 94 | 1.0000 | 1.0000 | 0.003301 |
| 143 | 94 | 63 | 1.0000 | 1.0000 | 0.018892 |
| 144 | 63 | 175 | 1.0000 | 1.0000 | 0.063402 |
| 145 | 175 | 85 | 1.0000 | 1.0000 | 0.113933 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 146 | 85 | 179 | 1.0000 | 1.0000 | 0.404269 |
| 147 | 181 | 138 | 1.0000 | 1.0000 | 0.162720 |
| 148 | 138 | 151 | 1.0000 | 1.0000 | 0.094954 |
| 149 | 151 | 37 | 1.0000 | 1.0000 | 0.046068 |
| 150 | 37 | 182 | 1.0000 | 1.0000 | 0.093076 |
| 151 | 182 | 180 | 1.0000 | 1.0000 | 0.080838 |
| 152 | 180 | 24 | 1.0000 | 1.0000 | 0.036574 |
| 153 | 24 | 111 | 1.0000 | 1.0000 | 0.025192 |
| 154 | 111 | 184 | 1.0000 | 1.0000 | 0.327090 |
| 155 | 184 | 183 | 1.0000 | 1.0000 | 0.156461 |
| 156 | 183 | 174 | 1.0000 | 1.0000 | 0.199705 |
| 157 | 174 | 10 | 1.0000 | 1.0000 | 0.105480 |
| 158 | 10 | 53 | 1.0000 | 1.0000 | 0.012496 |
| 159 | 53 | 127 | 1.0000 | 1.0000 | 0.004370 |
| 160 | 127 | 104 | 1.0000 | 1.0000 | 0.004632 |
| 161 | 98 | 68 | 1.0000 | 1.0000 | 0.009098 |
| 162 | 68 | 161 | 1.0000 | 1.0000 | 0.000679 |
| 163 | 161 | 193 | 1.0000 | 1.0000 | 0.146729 |
| 164 | 193 | 55 | 1.0000 | 1.0000 | 0.099068 |
| 165 | 55 | 196 | 1.0000 | 1.0000 | 0.409181 |
| 166 | 196 | 105 | 1.0000 | 1.0000 | 0.449373 |
| 167 | 105 | 113 | 1.0000 | 1.0000 | 0.021191 |
| 168 | 113 | 11 | 1.0000 | 1.0000 | 0.015352 |
| 169 | 11 | 195 | 1.0000 | 1.0000 | 0.026296 |
| 170 | 195 | 169 | 1.0000 | 1.0000 | 0.019695 |
| 171 | 169 | 27 | 1.0000 | 1.0000 | 0.072854 |
| 172 | 27 | 188 | 1.0000 | 1.0000 | 0.183846 |
| 173 | 188 | 182 | 1.0000 | 1.0000 | 0.152409 |
| 174 | 182 | 154 | 1.0000 | 1.0000 | 0.023407 |
| 175 | 154 | 40 | 1.0000 | 1.0000 | 0.128336 |
| 176 | 40 | 80 | 1.0000 | 1.0000 | 0.056927 |
| 177 | 80 | 130 | 1.0000 | 1.0000 | 0.084526 |
| 178 | 130 | 141 | 1.0000 | 1.0000 | 0.319736 |
| 179 | 162 | 194 | 1.0000 | 1.0000 | 0.043235 |
| 180 | 194 | 69 | 1.0000 | 1.0000 | 0.016731 |
| 181 | 69 | 143 | 1.0000 | 1.0000 | 0.120226 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 182 | 143 | 196 | 1.0000 | 1.0000 | 0.536841 |
| 183 | 196 | 106 | 1.0000 | 1.0000 | 0.449250 |
| 184 | 106 | 12 | 1.0000 | 1.0000 | 0.022800 |
| 185 | 12 | 171 | 1.0000 | 1.0000 | 0.241741 |
| 186 | 171 | 197 | 1.0000 | 1.0000 | 0.769690 |
| 187 | 197 | 184 | 1.0000 | 1.0000 | 0.774085 |
| 188 | 184 | 156 | 1.0000 | 1.0000 | 0.132746 |
| 189 | 156 | 132 | 1.0000 | 1.0000 | 0.212448 |
| 190 | 132 | 189 | 1.0000 | 1.0000 | 0.258146 |
| 191 | 189 | 82 | 1.0000 | 1.0000 | 0.289638 |
| 192 | 82 | 115 | 1.0000 | 1.0000 | 0.018140 |
| 193 | 115 | 28 | 1.0000 | 1.0000 | 0.021407 |
| 194 | 28 | 41 | 1.0000 | 1.0000 | 0.040850 |
| 195 | 16 | 20 | 1.0000 | 1.0000 | 0.020121 |
| 196 | 20 | 14 | 1.0000 | 1.0000 | 0.019197 |
| 197 | 14 | 26 | 1.0000 | 1.0000 | 0.019625 |
| 198 | 26 | 22 | 1.0000 | 1.0000 | 0.019225 |
| 199 | 22 | 13 | 1.0000 | 1.0000 | 0.005938 |
| 200 | 13 | 19 | 1.0000 | 1.0000 | 0.003262 |
| 201 | 19 | 27 | 1.0000 | 1.0000 | 0.008537 |
| 202 | 27 | 24 | 1.0000 | 1.0000 | 0.012826 |
| 203 | 24 | 25 | 1.0000 | 1.0000 | 0.020048 |
| 204 | 25 | 21 | 1.0000 | 1.0000 | 0.020926 |
| 205 | 21 | 18 | 1.0000 | 1.0000 | 0.017376 |
| 206 | 18 | 15 | 1.0000 | 1.0000 | 0.006735 |
| 207 | 15 | 28 | 1.0000 | 1.0000 | 0.001133 |
| 208 | 28 | 17 | 1.0000 | 1.0000 | 0.020576 |
| 209 | 17 | 23 | 1.0000 | 1.0000 | 0.019745 |
| 210 | 41 | 32 | 1.0000 | 1.0000 | 0.040850 |
| 211 | 32 | 30 | 1.0000 | 1.0000 | 0.016370 |
| 212 | 30 | 40 | 1.0000 | 1.0000 | 0.021803 |
| 213 | 40 | 35 | 1.0000 | 1.0000 | 0.035502 |
| 214 | 35 | 37 | 1.0000 | 1.0000 | 0.023166 |
| 215 | 37 | 38 | 1.0000 | 1.0000 | 0.029485 |
| 216 | 38 | 33 | 1.0000 | 1.0000 | 0.022012 |
| 217 | 33 | 13 | 1.0000 | 1.0000 | 0.021940 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 218 | 13 | 36 | 1.0000 | 1.0000 | 0.019264 |
| 219 | 36 | 39 | 1.0000 | 1.0000 | 0.031165 |
| 220 | 39 | 29 | 1.0000 | 1.0000 | 0.018709 |
| 221 | 29 | 34 | 1.0000 | 1.0000 | 0.024046 |
| 222 | 34 | 31 | 1.0000 | 1.0000 | 0.011042 |
| 223 | 95 | 94 | 1.0000 | 1.0000 | 0.004580 |
| 224 | 94 | 56 | 1.0000 | 1.0000 | 0.001494 |
| 225 | 56 | 98 | 1.0000 | 1.0000 | 0.006391 |
| 226 | 98 | 91 | 1.0000 | 1.0000 | 0.002707 |
| 227 | 91 | 44 | 1.0000 | 1.0000 | 0.012987 |
| 228 | 44 | 97 | 1.0000 | 1.0000 | 0.028897 |
| 229 | 97 | 93 | 1.0000 | 1.0000 | 0.019437 |
| 230 | 93 | 96 | 1.0000 | 1.0000 | 0.021995 |
| 231 | 96 | 92 | 1.0000 | 1.0000 | 0.011804 |
| 232 | 92 | 31 | 1.0000 | 1.0000 | 0.031163 |
| 233 | 31 | 16 | 1.0000 | 1.0000 | 0.020121 |
| 234 | 60 | 144 | 1.0000 | 1.0000 | 0.022838 |
| 235 | 144 | 146 | 1.0000 | 1.0000 | 0.163095 |
| 236 | 146 | 20 | 1.0000 | 1.0000 | 0.132970 |
| 237 | 20 | 34 | 1.0000 | 1.0000 | 0.029201 |
| 238 | 34 | 92 | 1.0000 | 1.0000 | 0.042205 |
| 239 | 92 | 148 | 1.0000 | 1.0000 | 0.124981 |
| 240 | 148 | 83 | 1.0000 | 1.0000 | 0.228195 |
| 241 | 83 | 147 | 1.0000 | 1.0000 | 0.286193 |
| 242 | 147 | 49 | 1.0000 | 1.0000 | 0.422431 |
| 243 | 49 | 145 | 1.0000 | 1.0000 | 0.086560 |
| 244 | 155 | 76 | 1.0000 | 1.0000 | 0.168154 |
| 245 | 76 | 150 | 1.0000 | 1.0000 | 0.065455 |
| 246 | 150 | 125 | 1.0000 | 1.0000 | 0.049534 |
| 247 | 125 | 156 | 1.0000 | 1.0000 | 0.191248 |
| 248 | 156 | 109 | 1.0000 | 1.0000 | 0.215749 |
| 249 | 109 | 152 | 1.0000 | 1.0000 | 0.068038 |
| 250 | 152 | 21 | 1.0000 | 1.0000 | 0.023277 |
| 251 | 21 | 154 | 1.0000 | 1.0000 | 0.061830 |
| 252 | 154 | 35 | 1.0000 | 1.0000 | 0.092834 |
| 253 | 35 | 151 | 1.0000 | 1.0000 | 0.069234 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 254 | 151 | 135 | 1.0000 | 1.0000 | 0.048243 |
| 255 | 135 | 149 | 1.0000 | 1.0000 | 0.016381 |
| 256 | 149 | 153 | 1.0000 | 1.0000 | 0.123549 |
| 257 | 153 | 61 | 1.0000 | 1.0000 | 0.026244 |
| 258 | 61 | 144 | 1.0000 | 1.0000 | 0.028548 |
| 259 | 176 | 78 | 1.0000 | 1.0000 | 0.072557 |
| 260 | 78 | 189 | 1.0000 | 1.0000 | 0.270723 |
| 261 | 189 | 112 | 1.0000 | 1.0000 | 0.286980 |
| 262 | 112 | 128 | 1.0000 | 1.0000 | 0.006003 |
| 263 | 128 | 152 | 1.0000 | 1.0000 | 0.036501 |
| 264 | 152 | 25 | 1.0000 | 1.0000 | 0.044203 |
| 265 | 25 | 180 | 1.0000 | 1.0000 | 0.016526 |
| 266 | 180 | 188 | 1.0000 | 1.0000 | 0.233246 |
| 267 | 188 | 38 | 1.0000 | 1.0000 | 0.215999 |
| 268 | 38 | 139 | 1.0000 | 1.0000 | 0.003407 |
| 269 | 139 | 166 | 1.0000 | 1.0000 | 0.026151 |
| 270 | 166 | 187 | 1.0000 | 1.0000 | 0.030932 |
| 271 | 187 | 65 | 1.0000 | 1.0000 | 0.056627 |
| 272 | 65 | 146 | 1.0000 | 1.0000 | 0.176091 |
| 273 | 129 | 79 | 1.0000 | 1.0000 | 0.051060 |
| 274 | 79 | 181 | 1.0000 | 1.0000 | 0.226770 |
| 275 | 181 | 168 | 1.0000 | 1.0000 | 0.003951 |
| 276 | 168 | 153 | 1.0000 | 1.0000 | 0.012792 |
| 277 | 153 | 187 | 1.0000 | 1.0000 | 0.067319 |
| 278 | 187 | 26 | 1.0000 | 1.0000 | 0.060926 |
| 279 | 26 | 39 | 1.0000 | 1.0000 | 0.025267 |
| 280 | 39 | 66 | 1.0000 | 1.0000 | 0.037723 |
| 281 | 66 | 96 | 1.0000 | 1.0000 | 0.059041 |
| 282 | 96 | 192 | 1.0000 | 1.0000 | 0.012739 |
| 283 | 192 | 88 | 1.0000 | 1.0000 | 0.091468 |
| 284 | 88 | 147 | 1.0000 | 1.0000 | 0.298989 |
| 285 | 49 | 43 | 1.0000 | 1.0000 | 0.092170 |
| 286 | 43 | 44 | 1.0000 | 1.0000 | 0.065150 |
| 287 | 44 | 48 | 1.0000 | 1.0000 | 0.023266 |
| 288 | 48 | 50 | 1.0000 | 1.0000 | 0.015340 |
| 289 | 50 | 54 | 1.0000 | 1.0000 | 0.013313 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 290 | 54 | 42 | 1.0000 | 1.0000 | 0.010153 |
| 291 | 42 | 55 | 1.0000 | 1.0000 | 0.011060 |
| 292 | 55 | 52 | 1.0000 | 1.0000 | 0.006749 |
| 293 | 52 | 46 | 1.0000 | 1.0000 | 0.008010 |
| 294 | 46 | 45 | 1.0000 | 1.0000 | 0.014696 |
| 295 | 45 | 51 | 1.0000 | 1.0000 | 0.012709 |
| 296 | 51 | 53 | 1.0000 | 1.0000 | 0.012761 |
| 297 | 53 | 47 | 1.0000 | 1.0000 | 0.004634 |
| 298 | 84 | 87 | 1.0000 | 1.0000 | 0.004686 |
| 299 | 87 | 90 | 1.0000 | 1.0000 | 0.004953 |
| 300 | 90 | 85 | 1.0000 | 1.0000 | 0.002290 |
| 301 | 85 | 86 | 1.0000 | 1.0000 | 0.007787 |
| 302 | 86 | 43 | 1.0000 | 1.0000 | 0.004252 |
| 303 | 43 | 88 | 1.0000 | 1.0000 | 0.031272 |
| 304 | 88 | 83 | 1.0000 | 1.0000 | 0.012797 |
| 305 | 83 | 89 | 1.0000 | 1.0000 | 0.020825 |
| 306 | 93 | 160 | 1.0000 | 1.0000 | 0.000054 |
| 307 | 160 | 162 | 1.0000 | 1.0000 | 0.092253 |
| 308 | 162 | 136 | 1.0000 | 1.0000 | 0.099060 |
| 309 | 136 | 50 | 1.0000 | 1.0000 | 0.002867 |
| 310 | 50 | 62 | 1.0000 | 1.0000 | 0.000841 |
| 311 | 62 | 161 | 1.0000 | 1.0000 | 0.012295 |
| 312 | 161 | 157 | 1.0000 | 1.0000 | 0.014361 |
| 313 | 157 | 158 | 1.0000 | 1.0000 | 0.034666 |
| 314 | 158 | 159 | 1.0000 | 1.0000 | 0.009370 |
| 315 | 159 | 116 | 1.0000 | 1.0000 | 0.021065 |
| 316 | 116 | 163 | 1.0000 | 1.0000 | 0.060203 |
| 317 | 163 | 133 | 1.0000 | 1.0000 | 0.036610 |
| 318 | 89 | 148 | 1.0000 | 1.0000 | 0.207370 |
| 319 | 148 | 192 | 1.0000 | 1.0000 | 0.149524 |
| 320 | 192 | 97 | 1.0000 | 1.0000 | 0.028693 |
| 321 | 97 | 160 | 1.0000 | 1.0000 | 0.019382 |
| 322 | 160 | 194 | 1.0000 | 1.0000 | 0.049018 |
| 323 | 194 | 140 | 1.0000 | 1.0000 | 0.041557 |
| 324 | 140 | 54 | 1.0000 | 1.0000 | 0.003821 |
| 325 | 54 | 67 | 1.0000 | 1.0000 | 0.000660 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 326 | 67 | 193 | 1.0000 | 1.0000 | 0.120941 |
| 327 | 193 | 177 | 1.0000 | 1.0000 | 0.249224 |
| 328 | 177 | 185 | 1.0000 | 1.0000 | 0.295052 |
| 329 | 185 | 122 | 1.0000 | 1.0000 | 0.171678 |
| 330 | 122 | 190 | 1.0000 | 1.0000 | 0.024871 |
| 331 | 165 | 120 | 1.0000 | 1.0000 | 0.003518 |
| 332 | 120 | 185 | 1.0000 | 1.0000 | 0.154000 |
| 333 | 185 | 158 | 1.0000 | 1.0000 | 0.172253 |
| 334 | 158 | 95 | 1.0000 | 1.0000 | 0.042547 |
| 335 | 95 | 64 | 1.0000 | 1.0000 | 0.023965 |
| 336 | 64 | 175 | 1.0000 | 1.0000 | 0.073055 |
| 337 | 175 | 86 | 1.0000 | 1.0000 | 0.121720 |
| 338 | 86 | 186 | 1.0000 | 1.0000 | 0.326812 |
| 339 | 186 | 172 | 1.0000 | 1.0000 | 0.098033 |
| 340 | 186 | 179 | 1.0000 | 1.0000 | 0.085244 |
| 341 | 179 | 90 | 1.0000 | 1.0000 | 0.401979 |
| 342 | 90 | 70 | 1.0000 | 1.0000 | 0.024943 |
| 343 | 70 | 191 | 1.0000 | 1.0000 | 0.087661 |
| 344 | 191 | 123 | 1.0000 | 1.0000 | 0.058066 |
| 345 | 123 | 163 | 1.0000 | 1.0000 | 0.080331 |
| 346 | 163 | 134 | 1.0000 | 1.0000 | 0.023148 |
| 347 | 155 | 81 | 1.0000 | 1.0000 | 0.144210 |
| 348 | 81 | 131 | 1.0000 | 1.0000 | 0.002150 |
| 349 | 131 | 178 | 1.0000 | 1.0000 | 0.088274 |
| 350 | 178 | 183 | 1.0000 | 1.0000 | 0.213792 |
| 351 | 183 | 197 | 1.0000 | 1.0000 | 0.617624 |
| 352 | 197 | 170 | 1.0000 | 1.0000 | 0.847925 |
| 353 | 170 | 114 | 1.0000 | 1.0000 | 0.354614 |
| 354 | 114 | 195 | 1.0000 | 1.0000 | 0.005524 |
| 355 | 195 | 142 | 1.0000 | 1.0000 | 0.086633 |
| 356 | 118 | 84 | 1.0000 | 1.0000 | 0.032088 |
| 357 | 84 | 173 | 1.0000 | 1.0000 | 0.028255 |
| 358 | 173 | 172 | 1.0000 | 1.0000 | 0.200087 |
| 359 | 172 | 145 | 1.0000 | 1.0000 | 0.230137 |
| 360 | 167 | 190 | 1.0000 | 1.0000 | 0.175467 |
| 361 | 190 | 121 | 1.0000 | 1.0000 | 0.008090 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 362 | 121 | 159 | 1.0000 | 1.0000 | 0.006837 |
| 363 | 159 | 191 | 1.0000 | 1.0000 | 0.016873 |
| 364 | 191 | 87 | 1.0000 | 1.0000 | 0.117557 |
| 365 | 87 | 173 | 1.0000 | 1.0000 | 0.023569 |

## 4. Impedance Calculation

| Quantity | Value |
|---|---|
| Applied voltage (V) | 1.00 V |
| Resistance per segment k | 1 Ohm |
| Calculated impedance R_eq | 3022.931699 Ohm |
| Total current I | -2.172081e-05 A |

## Impedance for different k (resistance per segment) values

| k (Ohm/segment) | R_equivalent (Ohm) |
|---|---|
| 1 | 3.022932 |
| 10 | 30.229317 |
| 100 | 302.293170 |
| 1000 | 3022.931699 |
| 10000 | 30229.316988 |